THE REDUCTION OF A COHEN RING MODULO A POWER OF THE RESIDUE CHARACTERISTIC IS FORMALLY SMOOTH OVER Z/P^N

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance, as required for this source: the first line of the pinned chapter file is \input{preamble}, and the native preamble was not staged with the source, so no preamble text lives inside the chapter. The retained preamble used here is the project's single Stacks-wide preamble (the same 217-line / 8116-byte file that the already published chow and etale-cohomology cards use), copied mechanically line for line from sources/stacks-chow/preamble.tex lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros). Not one character was changed, no package was added, no macro was defined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. The same-chapter pointers are located at their \label lines inside the pinned chapter file; the cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Editorial note. The staged candidate list records ext_refs = 0 for this label, but the proof actually references two same-chapter results, which the generator resolved and recorded as navigation stubs; the induction itself stays inside the counted span. This is the second instance today of that staged field being unreliable.

Editorial note. The counted proof contains no bibliography citation, so no bibliography excerpt is attached.

Allowed standard prerequisites (editorial).

- Characterisation of separable field extensions by formal smoothness. Same chapter, Proposition labelled proposition-characterize-separable-field-extensions; invoked at proof line 45135. Settles the case n = 1, since the residue field of a Cohen ring of characteristic p is separable over its prime field.
- Lifting formal smoothness along a square-zero ideal. Same chapter, Lemma labelled lemma-lift-formal-smoothness; invoked at proof lines 45138-45141. Performs the induction step, from level n to level n + 1, for the ideal generated by p^n.

Source reference. Original lemma 031L: sources/stacks-algebra/pinned-original-source.tex, source lines 38317–38317. Navigation pointer for \ref{lemma-lift-formal-smoothness} used inside the counted span (line 45139); the argument that consumes it is retained in the span.

Date: stacks-project:algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0322: sources/stacks-algebra/pinned-original-source.tex, source lines 44692–44692. Navigation pointer for \ref{proposition-characterize-separable-field-extensions} used inside the counted span (line 45136); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $p > 0$ be a prime. Let Λ be a Cohen ring with residue field of characteristic p. For every $n \geq 1$ the ring map*

$$\mathbf{Z}/p^n\mathbf{Z} \to \Lambda/p^n\Lambda$$

is formally smooth.

Proof. If $n = 1$, this follows from Proposition 0322. For general n we argue by induction on n. Namely, if $\mathbf{Z}/p^n\mathbf{Z} \to \Lambda/p^n\Lambda$ is formally smooth, then we can apply Lemma 031L to the ring map $\mathbf{Z}/p^{n+1}\mathbf{Z} \to \Lambda/p^{n+1}\Lambda$ and the ideal $I = (p^n) \subset \mathbf{Z}/p^{n+1}\mathbf{Z}$. □